CSC311: Introduction to Machine Learning

Project Report

Predicting Sports from Survey Responses

Submitted by

Group 63083

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Fall 2026

1 Executive Summary

This section should be at most 1/4 page (1 paragraph).

- Summarize the model families you explored (at least three).
- State your best-performing model and its projected accuracy.
- Provide a high-level justification of why this model outperformed the others.

2 Data Exploration

The dataset contains 870 survey responses from 290 students. Each student answered the 12 questions three times, once for each sport: soccer, basketball, and baseball. It is a 3-class classification problem. Each row has 12 features, including one numeric rating, three Likert-scale questions, three multi-select questions, and five free-text questions (one of which is a number). We do not use the unique ID as a feature, because it identifies the respondent rather than the sport. The classes are perfectly balanced, with 232 examples for each sport in the training set.

Before data exploration, we split the dataset (seed 42) into a training set of 696 examples (80%) and a test set of 174 examples (20%). We split by student ID rather than by row because a student's three responses share the same writing style; otherwise the model could recognize respondents instead of sports and overestimate test accuracy. The test set was set aside and not used for any exploration, preprocessing decision or tuning; hyperparameters will instead be tuned by 5-fold cross-validation grouped by student ID within the training set (Section 3).

Seven training responses from three students are completely empty; we drop them from training since they carry no information, and all statistics below are computed on the remaining 689 rows.

2.1 Numeric features

First, the only numeric feature is the rating, a score from 0 to 10. The overall distribution is roughly symmetric with mean 5.4, median 5, and standard deviation 2.4. It separates Baseball from the other two sports. Mean ratings are 4.0 for Baseball, 6.0 for Basketball and 6.2 for Soccer. 48% of Baseball responses fall between 0 and 3 compared to 12% for Basketball and 16% for Soccer. Second, it has no missing values, no outliers, and no inconsistencies.

The rating gives us a clear predictive signal to differentiate between Baseball and the other two sports, so we should keep it. Additionally, as the rating is numeric with a known range, it is reasonable to use it directly as a feature, divide by 10 to scale it to $[0, 1]$, or standardize it using training mean and standard deviation. We will compare these options on the validation set for each of our chosen models to see which one works best.

2.2 Likert features

First, the three Likert questions (strategy, injury, individual) each have five answers from "Strongly disagree" to "Strongly agree". Figure 1 shows that Baseball responses disagree more often on injury; Soccer responses disagree more often on individual. Second, strategy and individual each have one missing value. We handle this by imputing the training median values. Injury has no missing values. There are no outliers or inconsistencies.

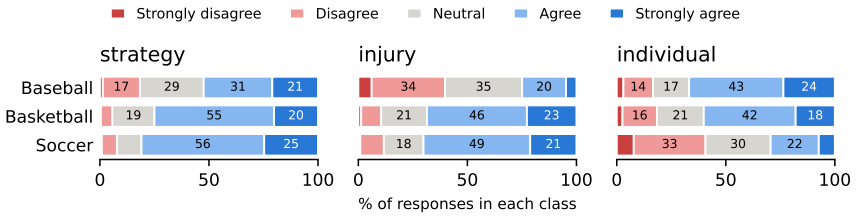


Figure 1: Distribution of Likert answers within each class in percent.

One way to encode the answers is as values from 1 to 5, preserving the natural order. Alternatively, one-hot encoding can be used and we will compare during validation. Encoding answers from 1 to 5 assumes equal gaps between agreement levels, while one-hot encoding creates more features. Also notice that the strategy question shows some separation between the sports but shows less separation than the other two questions. It is worth comparing models with/without this feature. For any missing values in the test set, we will impute the training median value as above.

2.3 Multi-select features

First, the three multi-select questions have 7 options for place, 4 for season, and 10 for music. Figure 2 shows that Indoor, Winter and Rap point to Basketball; Latin points to Soccer; Country points to Baseball, and Baseball is rarely played at home. Second, there are no missing values, no outliers, and no inconsistencies.

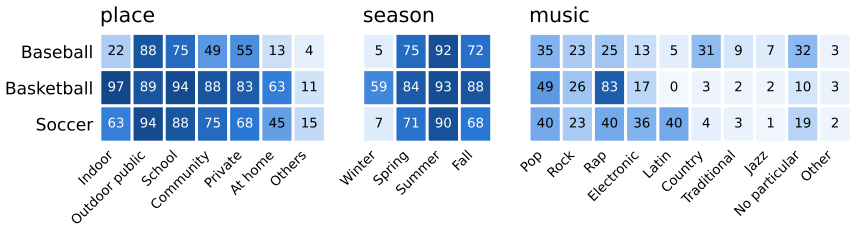


Figure 2: Percentage of each class selecting each multi-select option.

One way to represent these answers is using multi-hot encoding for each option, giving 21 features. We can also consider adding features for certain combinations of options. However, it is unclear which combinations are useful, or if we are just adding redundancy. To start, we will use the simple multi-hot encoding approach.

2.4 Free-text features

First, the four free-text questions (food, colour, song, feeling) are open-ended short answers. Each class has distinctive words, e.g., “waka waka” for Soccer and “orange” for Basketball. Second, we handle inconsistencies like “blue.” or “Blue” by standardizing capitalization and removing punctuation. We handle missing values like “none” or “idk” by adding a missing indicator. There are no outliers.

We will start with TF-IDF, keeping only words found in at least three training answers. Each word gets a feature based on its count in an answer, with less weight given to words appearing in many training answers. So less common words are weighted higher, which may help highlight the class distinctive words we observed in exploration. Any words that aren’t seen in the training set will be ignored at prediction time. This is reasonable since unseen words can’t be associated with any class using TF-IDF. Alternatively, we could use word embeddings to capture semantic similarity. We can compare these approaches during validation.

2.5 Price features

First, the price question asks for a number but is free text. The distribution is highly right-skewed: median 100, 95th percentile 1000. The median differs by class: 50 for Baseball, 150 for Basketball and 200 for Soccer. Second, there are some outliers up to 10000. We plan to keep these values and apply a $\log(1 + x)$ transformation. There are also inconsistencies like “5k”, “100-300” and “150CAD”. We handle these by parsing the text to float using regex. We handle missing values by imputing the training median value.

The differences in median price suggest it provides useful information for distinguishing the sports. We will keep the price feature and compare the performance of the model with and without the $\log$ transformation. For models sensitive to scale, such as regularized softmax regression, we will standardize prices after the selected transformation, using the training mean and standard deviation.

3 Methodology

3.1 Models to compare

We plan to compare the following model families:

- **Linear models:** Softmax regression is a simple linear model that can handle multi-class classification serving as a baseline. Above, we described methods to encode all the features into numeric values, so they can be used directly in a linear model. It isn’t clear whether the classes are linearly separable from our data exploration step, but from our data exploration, we observed that several features that have clear differences between classes. We can test whether a linear model is sufficient to capture these relationships.

- **Tree-based models:** Decision trees are a natural choice for classification problems, and they are universal function approximators allowing it to learn non-linear relationships. They don't require feature scaling and they are easily interpretable. They can also learn combinations of the Likert or multi-select features naturally, compared to e.g., softmax regression which can only add individual contributions of each feature. In our data exploration, we can see several features like "injury" which can clearly separate baseball from the other two sports, and a decision tree can learn to split on this feature. We can also consider ensemble methods like random forests to improve performance and reduce overfitting, which can be a disadvantage of standard decision trees.
- **Neural networks:** An MLP can learn non-linear relationships unlike pure linear models. It isn't as interpretable as a decision tree, but it can also learn complex relationships, we can use a final softmax layer to output the probabilities of each class, and it is worth testing.

3.2 Validation and preprocessing

We have split the dataset into a training set and a test set (80/20 split) by student ID. We plan to perform 5-fold cross-validation grouped by student ID within the training set as described in Section 2. The dataset is small, so the computational cost of cross-validation should be reasonable, and it will reduce the dependence of our results on a single random split. To be fair, we will use the same splits for all model families and feature setting.

3.3 Training and tuning

We plan to use scikit-learn and compare each setting using mean accuracy across the same five grouped validation folds. Tune all three models before selecting a family, and use seed 42 where applicable.

- **Softmax regression:** we plan to use L-BFGS to fit the model with L2 regularization. TODO: which hyperparameters to tune, what values to try, when to stop, etc.
- **Decision tree:** use Gini impurity to choose splits that make the resulting groups more concentrated in one sport. TODO
- **MLP:** start with one hidden layer using ReLU and train with Adam on cross-entropy loss. TODO

3.4 Evaluation and final model

We will tune all three model families and compare their average accuracy and F1 across the validation folds. If the performance is similar between models, we will prefer the simpler model. The selected model will be retrained on the entire training set and evaluated on the test set. We won't do any additional training or tuning after we have selected the final model and evaluated on the test set.

4 Results

This section should be at most 1 page.

The goal of the Results step is to show how well your models performed, explain which model you chose, and predict how it will do on the unseen test set.

Report the results of every model using the evaluation metrics you described in the Methodology section. Present them in tables or plots that make the comparison easy to read.

Compare performance across the three model families, and state clearly which model you chose as your final model. This must be the model implemented in your prediction script.

Analyze the errors your models make. Describe the common misclassifications or weaknesses you observed, and include a confusion matrix or a few illustrative examples if they help.

Estimate your final model's accuracy on the unseen test set as a single number. A range will earn no marks. Make sure to support your estimate with empirical evidence, such as validation results, the stability of your cross-validation scores, learning curves, or consistency across random seeds or folds.

5 Contributions and Learning

This section should contain a list of 3–4 bullets, one for each student. Each student should write at most three sentences describing their main contributions and the most important lesson learned.

6 References

Use references whenever appropriate (e.g., when you consult textbooks, research papers, online tutorials, or external code). Cite in the text using $\backslash\mathrm{cite}\{\}$. For example, let's cite the two references provided in the $.\mathrm{bib}$ file [1, 2].

References

[1] Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, 2006.

[2] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, 2009.

A Guiding Questions for Data Exploration

These questions support Section 2. Work through them in order, and record your answers for our dataset in that section rather than here. This appendix is part of the template instructions, so comment it out before you submit, as you do with the rest of the instructions.

1. Determine Input to Model
   (a) For each example, what features do we have? What are the types of the features (numerical, categorical, or text)?
   (b) In the released data set, each row has a unique ID identifying which student provided the data. Why should we not use this unique ID as a feature?

2. Data Splitting
   (a) Why is it a good idea to split the provided data into a training set and a test set?
   (b) Why is it a good idea to do the training and test set splitting based on the unique ID identifying each respondent?
   (c) What is the role of the validation set and why do we need this third split?
   (d) Why should we only perform data exploration on the training set and not on the validation or the test sets?

3. Feature Distributions
   One reason for exploring the distributions of features is to get to know the data before building any model. This exploration can help us catch common issues such as outliers, skewed distributions, class imbalance, missing values, and large difference in feature scales.
   (a) What should we do when exploring the distributions of the features?
   (b) Suppose a feature takes the same value for almost every respondent. Why is this feature worth finding, and what would you do about it?
   (c) Suppose two features are highly correlated, such as two Likert questions that respondents tend to answer the same way. How would you test whether two features are highly correlated, and why does a highly correlated pair cause trouble?

4. Class Imbalance
   (a) What is class imbalance? Why is class imbalance problematic?
   (b) Does the dataset have a class imbalance issue? If so, how severe?
   (c) What are some common strategies to address class imbalance?

5. Missing Values
   (a) First, determine if the dataset contains missing values. Report the percentage of values missing for each feature in a table. A missing value can take many forms

such as a null value, an empty string, an empty list, “n/a”, or “-”. Normalize all of these to a single missing marker before you count, or you will underestimate how much is missing.

(b) Next, for each feature with a substantial missing rate, try to determine why the values are likely missing. Different explanations may require different fixes. What are some possible explanations for missing values?

(c) Since you never see the missing value, it may not be possible to verify which explanation holds. However, it would be useful to determine whether the missingness of a feature is related to the label or not. If the missingness is related to the label, a missing value itself provides a predictive signal. What is a common strategy for determining this?

(d) What are some common strategies to handle missing values? One is called imputation.

(e) When is it appropriate to use each strategy above to handle missing values?

(f) Suppose that you decide to impute the median value for a numeric feature. Why is it problematic to do the imputation on the validation and test sets?

(g) Finally, decide how missing values should be handled for each feature in our dataset, and record your decision.

6. Outliers
   (a) What observation might cause you to suspect that certain values are outliers?
   (b) How do I determine whether a feature value should be considered an outlier?
   (c) What are some strategies to handle outliers?

7. Feature Engineering
   (a) How can we convert the responses to a multi-select question into a feature?
   (b) The survey has a few Likert-scale questions that ask respondents to indicate their level of agreement to a statement using an ordered set of options (e.g., from Strongly Disagree to Strongly Agree). We can convert the Likert-scale responses to features by modeling them as ordinal values or one-hot encoded categories. What are some advantages and disadvantages of each approach?
   (c) At least one free-text question asks for a number. How would you turn such a text response into a number, and when would you declare a response unparseable?
   (d) What are some strategies to convert the free-text responses to numeric features?
   (e) One common technique for processing the free-text responses is TF-IDF vectorization. Explain this technique in detail. Why is this technique suitable for processing the free-text data?

(f) How should we handle a word that appears at prediction time but never in the training split?

(g) After encoding all the features, you recorded that there are 1100 features and 300 examples. How would the ratio between these two inform your model choices? Why would you choose a simpler model with strong regularization over a larger model?

8. Feature Scaling

(a) Two common choices for feature scaling are standardization and min-max scaling. Explain each in one sentences. Why do we need to perform feature scaling for our dataset, which has Likert-scale, multi-hot encoded, and word count features?

(b) Is each model below affected by feature scaling? Briefly explain.

- K-nearest neighbours
- Decision trees and random forest
- Softmax regression trained using gradient descent
- L2 Regularization
- Naive Bayes

(c) You decide to standardize a feature. Which split do you use to compute the mean and standard deviation, and why?

9. Which Features Are Predictive

Next, we would like to determine whether a feature is useful by looking at the feature together with the label. This analysis could also tell you which model families are worth trying and help you justify your model choices in the Methodology section.

(a) How do you examine the relationship between a Likert-scale or numeric feature and the label?

(b) How do you examine the relationship between a multi-select or free-text feature and the label?

(c) Why should you only do this analysis on the training split and not on the validation or test splits?

(d) How can you delete whether a feature is worth keeping by comparing the difference in mean feature values between classes with the spread of feature values within a class?

Consider a Likert-scale feature on a scale of 1 to 5. Is the feature worth keeping if two classes have means 3.5 and 3.8 while the standard deviation of answers within a class is 1.2? What about if two classes have means 2.1 and 4.3 while the standard deviation of answers within a class is 0.6?

(e) What does the shape of the relationship tell you about which model families to try? Consider the three scenarios below. Discuss the model families you would try for each one.

i. The classes separate well along one feature at a threshold.
ii. The middle values of a Likert feature belong to one class while both extremes belong to another.
iii. After vectorizing the free-text responses, you have several hundred word-count features. No single word separates the classes, but each word in a response carries a small amount of evidence.

(f) Finally, write down the features you expect to be the most predictive, and name the plot or the number that led you to each one.

B Guiding Questions for Methodology

These questions support Section 3. Work through them in order, and record your answers for our dataset in that section rather than here. This appendix is part of the template instructions, so comment it out before you submit, as you do with the rest of the instructions.

1. Choosing Model Families
   (a) Based on the size of the data set, should we consider fine-tuning a large language model or starting with simpler models? What are some advantages and disadvantages of each approach?
   (b) Which findings from your data exploration would support choosing softmax regression? Which findings would support choosing a decision tree instead?

2. Training Details
   (a) Suppose you train a multilayer perceptron. Which choices would another student need to know to reproduce your results exactly?
   (b) Your softmax regression model reaches 95% accuracy on the training set but only 70% on the validation set. What does this gap tell you, and which regularization or early stopping strategies could you try?

3. Validation
   (a) What is cross validation? If we want to use cross validation, why should we use grouped cross validation, grouped by the unique ID?
   (b) Pre-processing steps such as normalization and text vectorization learn values from the data, like a mean or a vocabulary. Which part of the data should these values be computed from during cross validation, and why?

4. Hyperparameter Tuning
   (a) Which hyperparameters would you tune for a decision tree, and how would you choose the range of values to try?
   (b) You try maximum depths from 2 to 10 for a decision tree, and the validation accuracy is highest at a depth of 10. Is your search finished? What would you do next?

(c) After choosing your final model and its hyperparameters, should you retrain it on the training and validation sets combined? What do you gain, and what do you lose?

5. Comparing Model Families Fairly
   (a) With default hyperparameters, softmax regression reaches 78% validation accuracy and a decision tree reaches 72%. You tune softmax regression up to 82% and choose it as your final model. What is wrong with this process, and how would you fix it?

   (b) Across five folds, model A has a mean validation accuracy 1% higher than model B, but the accuracy of each model varies by about 3% from fold to fold. Can you conclude that model A is better? What else could help you decide?

6. Evaluation Metrics
   (a) Suppose one class makes up 60% of the training set. What accuracy does a model that always predicts this class achieve? Which metric would reveal that this model is useless?

   (b) How do you compute precision, recall, and F1 for a 3-class problem? What is the difference between macro-averaging and weighted averaging, and when would you prefer each one?

7. Additional Techniques
   (a) Suppose you apply PCA to several hundred word-count features. How would you decide how many components to keep, and which part of the data should you fit PCA on?

   (b) How could you use a clustering method such as k-means to explore the free-text responses? How would you tell whether the clusters improve your predictions?